

Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (GPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

Bachelor of Technology in Computer Science (GPA: 3.64/4)

May 2022

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C

Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, PHP, REST APIs

Automation and Testing: Selenium, PyAutoGUI

Databases & Tools: MySQL, PostgreSQL, Google Firebase, MariaDB, MySQL Workbench

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Designed onboarding solutions for content creators, optimized video ingestion and distribution, and used Terraform-based automation for customer onboarding, reducing manual setup time by 90%.
- Managed 15+ containerized Microservices on AWS EKS with Docker & Kubernetes, increasing deployment frequency by 30%. Worked extensively in Linux environments to deploy, monitor, and troubleshoot systems.
- Deployed GitOps-based pipelines via ArgoCD, enhancing CI/CD reliability and rollback efficiency. Developed Python scripts and integrated APIs with partner systems for video preprocessing and verification before CDN delivery, reducing failures by 95%.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- Optimized MySQL database using PHP and stored procedures, improving query speed by 50% and enabling faster data access for over 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, and JavaScript, reducing search time by 60% and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Plethora - A Fest Management System (MERN Stack)

- Built a full-stack event platform for event discovery, registration, and authentication.
- Enhanced MariaDB database performance through in-depth query analysis and index optimization, decreasing real-time query response time by 50%.

Expense Tracker (Kotlin, Java, and Google Firebase)

- Developed a mobile expense tracker application using Android Studio; enabled real-time budget monitoring.
- Improved Firebase synchronization efficiency by 33%, resulting in faster data updates and a smoother user experience.

Handling Imbalanced Data on Bank Churn Prediction (Machine Learning & Data Analytics)

- Developed an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed 100+ real-time autograder test cases